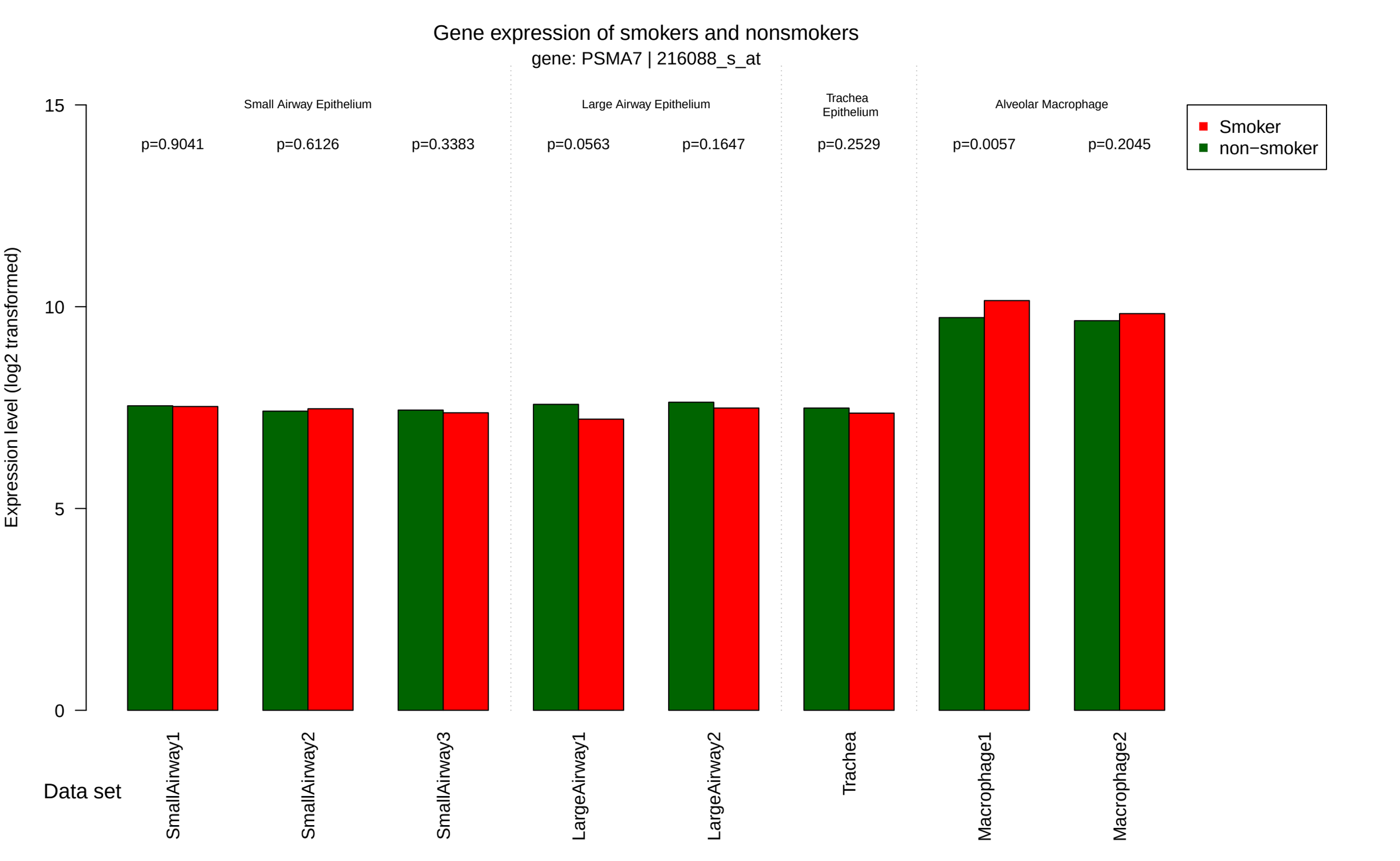
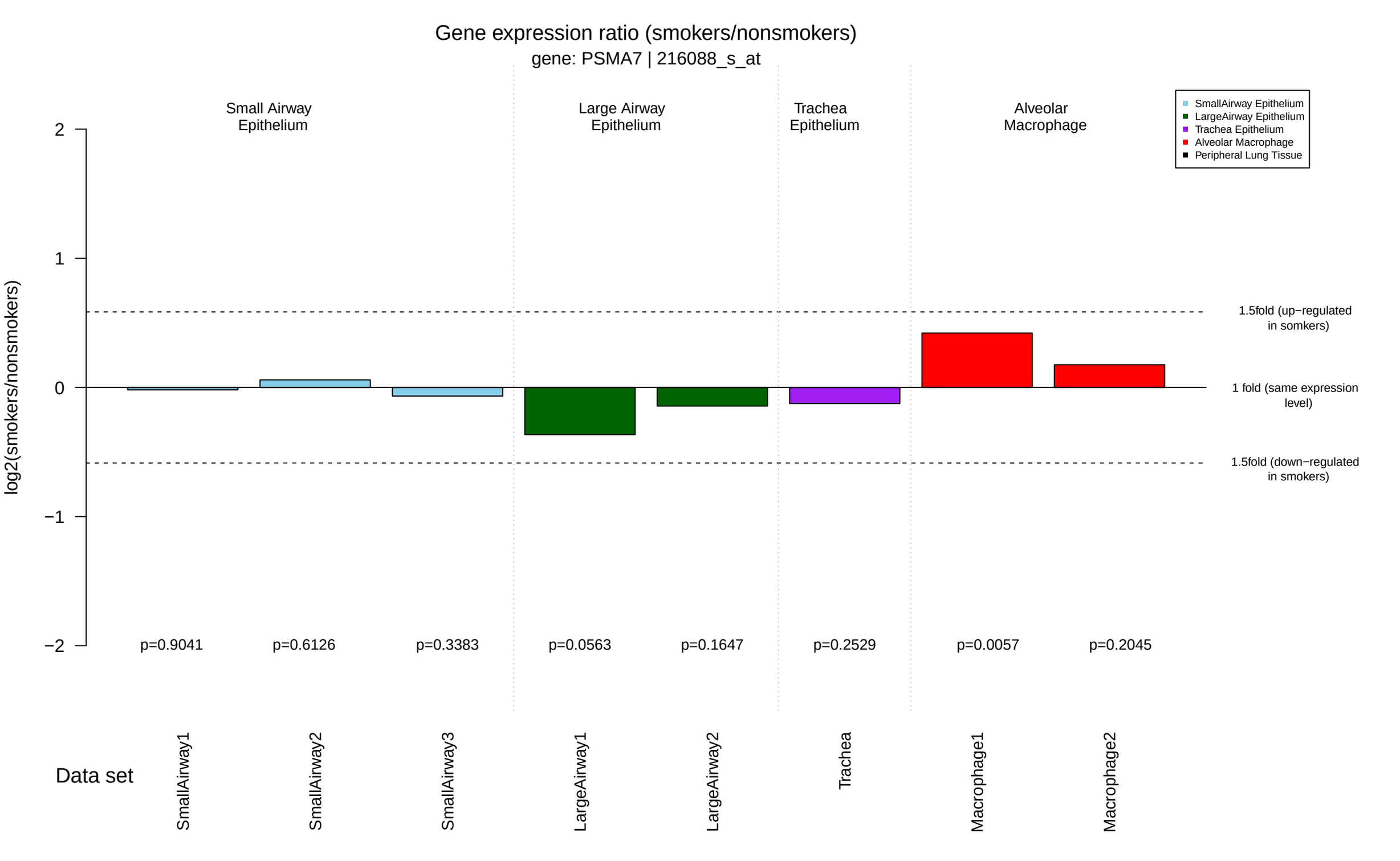
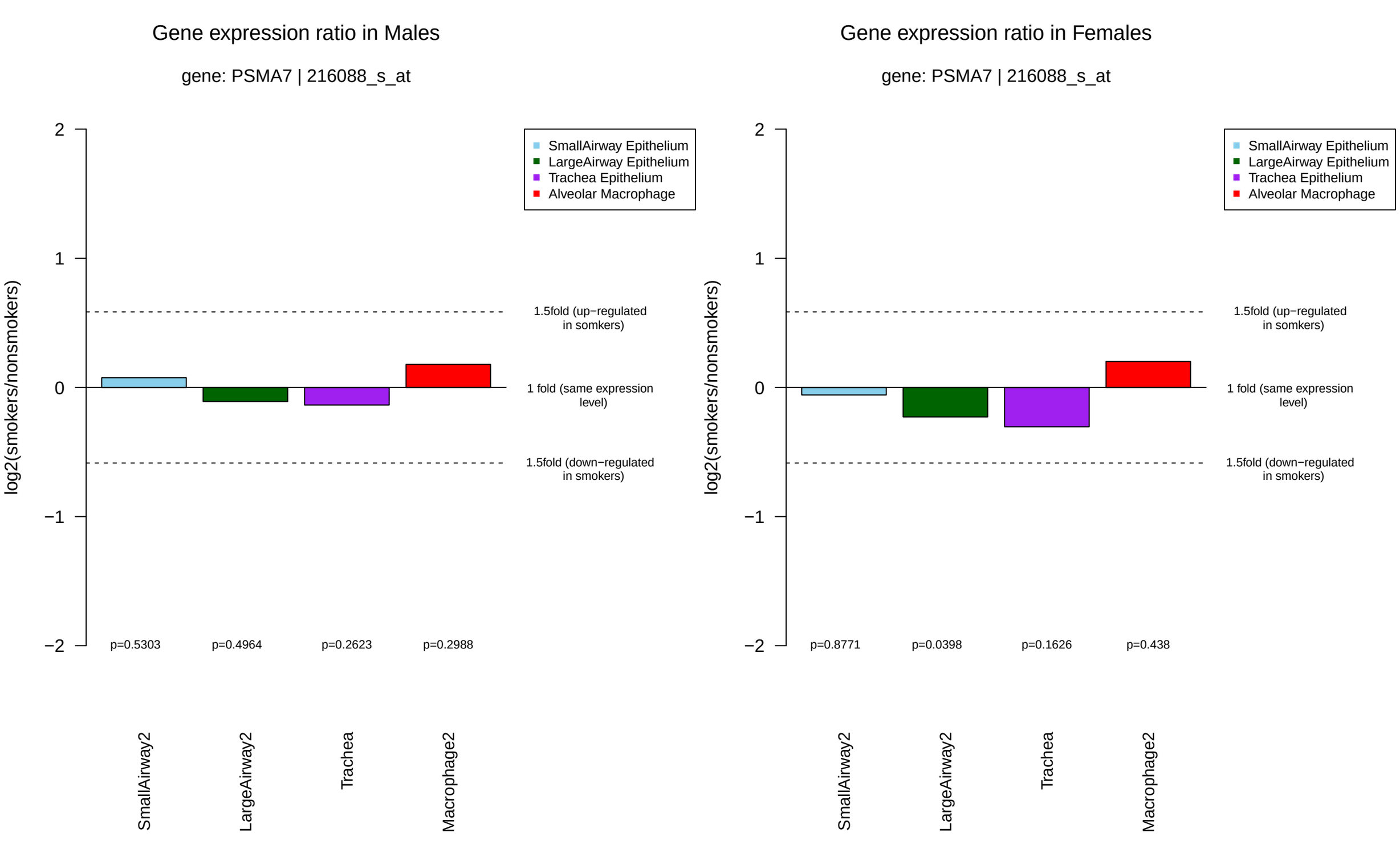
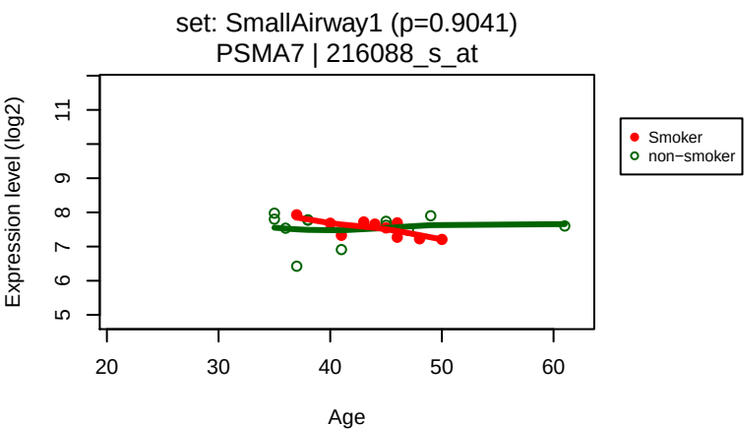
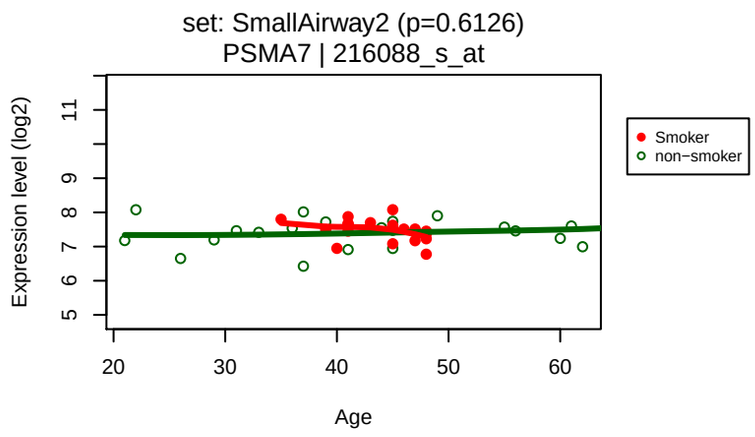
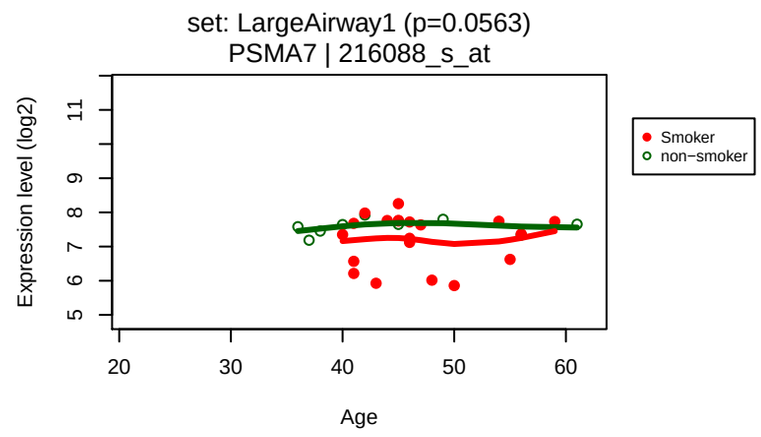

Gene expression of smokers and nonsmokers
gene: PSMA7 | 216088_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9041
p=0.6126
p=0.3383
p=0.0563
p=0.1647
p=0.2529
p=0.0057
p=0.2045
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

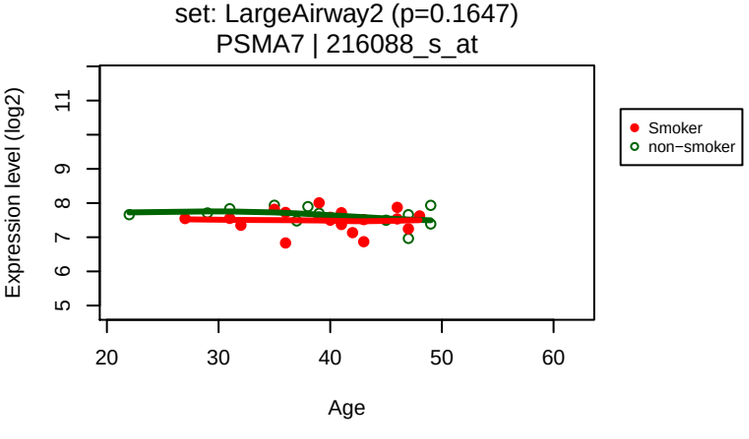

Gene expression ratio (smokers/nonsmokers)
gene: PSMA7 | 216088_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9041
p=0.6126
p=0.3383
p=0.0563
p=0.1647
p=0.2529
p=0.0057
p=0.2045
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

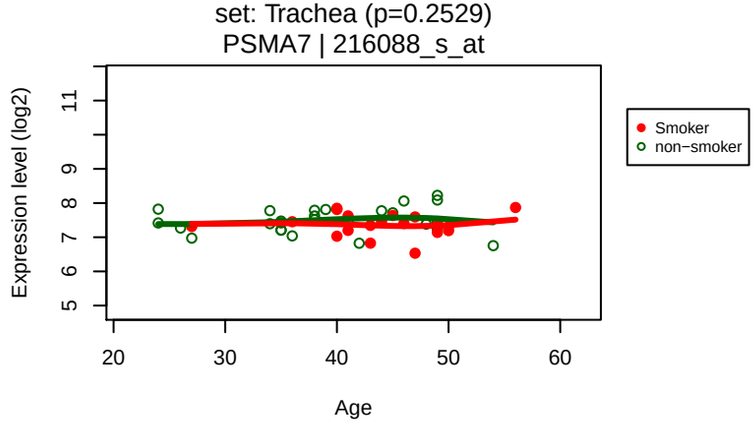

Gene expression ratio in Males
gene: PSMA7 | 216088_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5303
p=0.4964
p=0.2623
p=0.2988
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA7 | 216088_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8771
p=0.0398
p=0.1626
p=0.438
SmallAirway2
LargeAirway2
Trachea
Macrophage2

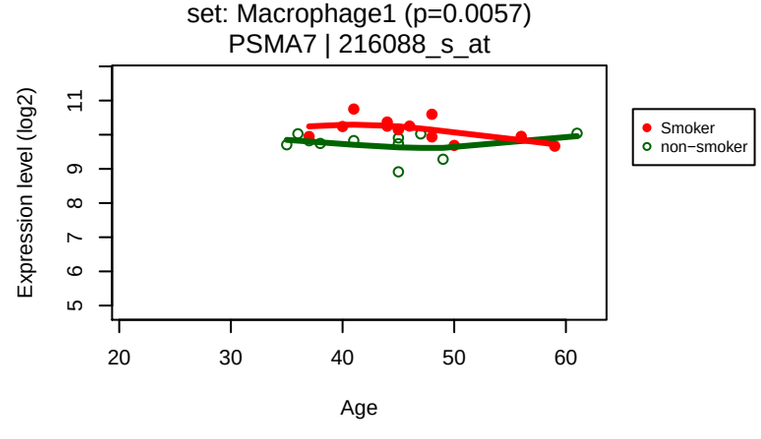

set: SmallAirway1 (p=0.9041)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker

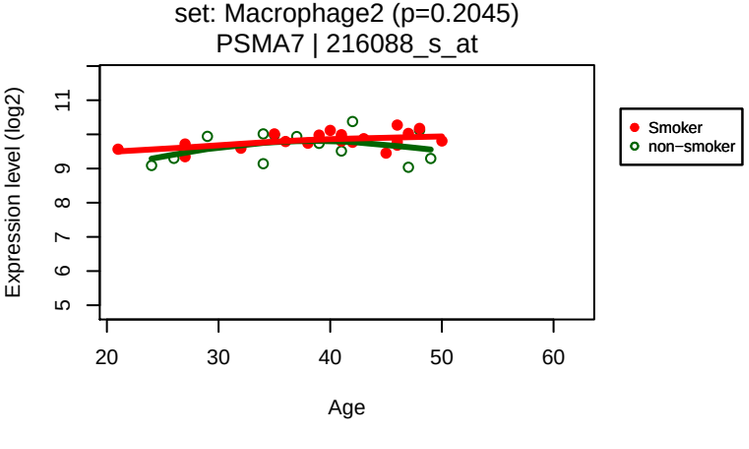

set: SmallAirway2 (p=0.6126)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker

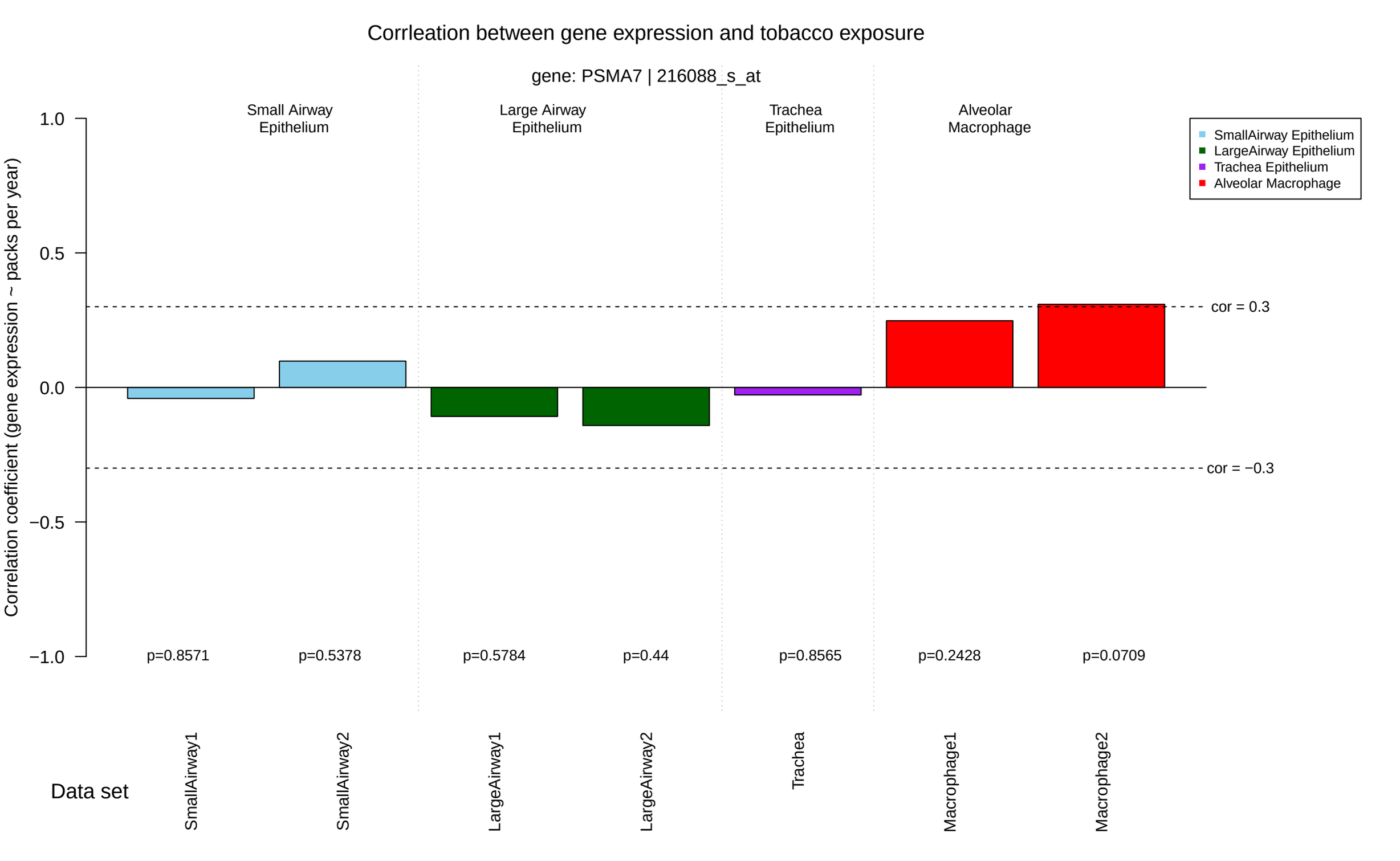

set: LargeAirway1 (p=0.0563)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1647)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2529)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2045)
PSMA7 | 216088_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA7 | 216088_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8571
p=0.5378
p=0.5784
p=0.44
p=0.8565
p=0.2428
p=0.0709
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2